

The Non-Normal Grid: A Resolvent Framework for Hidden Transient Fragility and Non-Modal Weak Nodes in Inverter-Dominated Networks

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Abstract

Abstract—Dynamic security of inverter-dominated grids is routinely certified through the eigenvalues of an inertia-weighted Laplacian: a grid with critical eigenvalue $\nu_c > 0$ and all poles damped is declared safe. We show this criterion is systematically optimistic. We build the network as a multi-agent system—local second-order node dynamics coupled through a mass-normalized graph operator—and show that grid-following control inserts an asymmetric term that renders the global operator *non-normal*. For non-normal operators the eigenvalues do not bound the transient response, so a grid can pass every eigenvalue and voltage test while amplifying a small disturbance past protection thresholds. We diagnose this with the *resolvent operator* and its singular value decomposition, obtaining: (i) a frequency-resolved amplification curve revealing a hidden resonance; (ii) a Non-Modal Weakness Index whose optimal-forcing and optimal-response nodes differ and disagree with short-circuit ratio; and (iii) a transient-aware hosting certificate that lower-bounds the eigenvalue hosting limit, together with a resolvent reading of the Dynamic Braess Effect. The framework is closed-form, simulation-free, and demonstrated on the IEEE 39-bus system.

Index Terms—Non-normal operators, resolvent analysis, pseudospectrum, dynamic hosting capacity, inverter-based resources, weak nodes, multi-agent systems.

I. INTRODUCTION

The displacement of synchronous generators (SGs) by inverter-based resources (IBRs) reshapes the dynamical structure of power networks. The prevailing analytical response models the low-inertia grid as a networked second-order system and certifies stability through the spectrum of a mass-normalized graph Laplacian, with the smallest nonzero eigenvalue ν_c as the binding margin on dynamic hosting capacity. This spectral viewpoint is accurate for high-inertia systems and underpins much recent planning work.

This paper identifies a structural blind spot. Eigenvalues govern the response of a linear system *only when its operator is normal*—when the eigenvectors are orthogonal. Grid-following (GFL) control, through its frequency-to-angle feedback, destroys normality, and the departure grows with IBR penetration. As a consequence a grid can satisfy every eigenvalue-based and voltage-based test while still amplifying a modest disturbance enough to violate frequency limits and trip protection—a failure mode invisible to current hosting screening.

We organize the analysis as a five-stage multi-agent framework: local agent dynamics (§II); mass-normalized coupling and the asymmetric control term (§III); global assembly and the precise origin of non-normality (§IV); resolvent certification (§V); and the deliverables—amplification curve, non-modal weak nodes, hosting certificate, and Braess effect (§VI). Throughout we separate reusable tools from the contribution.

II. STAGE 1: LOCAL AGENT DYNAMICS

Each bus i carries phase θ_i and frequency deviation ω_i , with state $x_i = [\theta_i, \omega_i]^\top$. The isolated dynamics are

$$\dot{x}_i = A_{\text{loc},i} x_i, \quad A_{\text{loc},i} = \begin{bmatrix} 0 & 1 \\ 0 & -\gamma_i \end{bmatrix}, \quad \gamma_i = \frac{D_i}{M_i}, \quad (1)$$

with effective inertia $M_i > 0$ and damping $D_i \geq 0$. This unifies device classes: SGs have large M_i ; GFMs synthesize M_i, D_i by control; GFL units have $M_i = \epsilon \ll M_{SG}$ with D_i set by the PLL gain via a standard singular-perturbation argument. The second-order structure is thus preserved network-wide.

III. STAGE 2: NETWORK COUPLING

With stiff voltages and high X/R , the linearized active-power Jacobian is the weighted Laplacian, $\partial P_e / \partial \theta|_{\theta^*} = L = L^\top \succeq 0$, $L\mathbf{1} = 0$. The mass-normalized operator is

$$\tilde{L} = M^{-1/2} L M^{-1/2} = \tilde{L}^\top \succeq 0, \quad 0 = \nu_1 < \nu_2 \leq \dots, \quad (2)$$

with $\nu_c := \nu_2$. So far the operator is symmetric. The contribution begins here: GFL control does not realize a symmetric restoring force. Its PLL measures frequency and feeds it into the angle channel through a finite-bandwidth, directional gain, contributing an *asymmetric* perturbation Δ to the Jacobian, $J = L + \Delta$, $\Delta \neq \Delta^\top$. This asymmetry, not the loss of inertia alone, is the structural agent of fragility.

IV. STAGE 3: GLOBAL ASSEMBLY AND THE ORIGIN OF NON-NORMALITY

Stacking agents (node-major) and coupling through the Laplacian gives the Kronecker form

$$\dot{X} = [I_n \otimes A_{\text{loc}} - (M^{-1}L) \otimes \Gamma]X + (I_n \otimes B_0)u, \quad (3)$$

$\Gamma = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $B_0 = [0, 1]^\top$. The Kronecker structure cleanly separates device physics (A_{loc}) from topology ($M^{-1}L$); we use it for assembly only. Re-ordering to state-major $x = [\theta^\top, \omega^\top]^\top$,

$$\dot{x} = A_{\text{sys}}x + Bu, \quad A_{\text{sys}} = \begin{bmatrix} 0 & I \\ -M^{-1}L & -M^{-1}D \end{bmatrix} + \Delta_{\text{glob}}, \quad (4)$$

where Δ_{glob} is the global image of the GFL term.

Proposition 1 (Origin of non-normality). *With heterogeneous GFL units the global operator satisfies $A_{\text{sys}}^\top A_{\text{sys}} \neq A_{\text{sys}} A_{\text{sys}}^\top$; equivalently, its eigenvectors are non-orthogonal. The departure from normality, measured by Henrici's index $d_F(A_{\text{sys}}) = (\|A_{\text{sys}}\|_F^2 - \sum_k |\lambda_k|^2)^{1/2} \geq 0$, is strictly positive and increases with IBR penetration ρ .*

Remark 1 (What invalidates the eigenvalue certificate). Normality—not commutation of the Kronecker factors—is the relevant property. For a normal operator the modal expansion is orthogonal, $e^{A_{\text{sys}}t} = \sum_k e^{\lambda_k t} u_k u_k^*$, giving $\|e^{A_{\text{sys}}t}\| = e^{t\alpha(A_{\text{sys}})}$, $\alpha = \max_k \Re \lambda_k$: eigenvalues govern the response. When normality fails, the expansion uses *oblique* projectors $v_k \tilde{v}_k^*$ with $\|v_k \tilde{v}_k^*\| \gg 1$, and only the conditioned bound $\|e^{A_{\text{sys}}t}\| \leq \kappa(V)e^{t\alpha}$ holds, active for intermediate t . The factor $\kappa(V)$ is precisely the hidden transient amplification.

V. STAGE 4: RESOLVENT CERTIFICATION

Force the system at frequency Ω , $u = \hat{u}e^{i\Omega t}$; the steady response is $\hat{x} = (\Omega)\hat{u}$ with the resolvent

$$(\Omega) = (i\Omega I - A_{\text{sys}})^{-1}B. \quad (5)$$

With output selector $C = [0 \ I]$, form $H(\Omega) = C(\Omega)$ and its SVD $H(\Omega) = \sum_j \sigma_j(\Omega) \phi_j(\Omega) \psi_j(\Omega)^*$. Here σ_1 is the worst-case gain at Ω ; ψ_1 the optimal forcing (most dangerous spatial disturbance); ϕ_1 the optimal response (where the amplified swing appears).

Proposition 2 (Normal vs. non-normal gain). *If A_{sys} is normal, $\sigma_{\max}(H(\Omega)) = \max_k |i\Omega - \lambda_k|^{-1}$, peaking only at eigenfrequencies. If non-normal, $\sigma_{\max}(H(\Omega)) \leq \kappa(V) \max_k |i\Omega - \lambda_k|^{-1}$, with the excess $\kappa(V)$ producing gain no eigenvalue predicts.*

The resolvent unifies the relevant margins: the peak $\gamma_\infty = \sup_\Omega \sigma_{\max}(H(\Omega))$ is the H_∞ worst-case gain (transient); the integral $\frac{1}{2\pi} \int \|H(\Omega)\|_F^2 d\Omega \propto \sum_{k \geq 2} \nu_k^{-1}$ is the H_2

stochastic variance; and the level set of $\|(zI - A_{\text{sys}})^{-1}\|$ in $\mathbb{C}$ is the pseudospectrum. One operator certifies transient, stochastic, and spectral fragility.

VI. STAGE 5: DELIVERABLES

A. Hidden amplification curve

Sweeping penetration ρ and plotting $\sigma_1(\Omega; \rho)$ yields a flat, low curve at high inertia and a sharp resonance near $\Omega^* \approx \sqrt{\nu_c}$ as $\rho \rightarrow 1$ —of height $\propto \kappa(V)$, far above the eigenvalue prediction. The grid is eigenvalue-stable yet has a large-gain window where a modest disturbance exceeds protection limits.

B. Non-Modal Weak Nodes

Definition 1. At the peak Ω^* , $W_i^{\text{nm}} = |[\psi_1(\Omega^*)]_i|^2$ (input) and $W_j^{\text{out}} = |[\phi_1(\Omega^*)]_j|^2$ (output).

Proposition 3. *For non-normal A_{sys} , $\psi_1 \neq \phi_1$ in general, and neither aligns with the critical eigenvector or the SCR ranking: the most dangerous injection node and the collapse node differ.*

C. Transient-aware hosting certificate

With protection margin Γ ,

$$\rho_{\text{io}}^* = \max\{\rho : \sup_\Omega \sigma_{\max}(H(\Omega; \rho)) \leq \Gamma\} \leq \rho_{\text{eig}}^*, \quad (6)$$

computed by a 1-D sweep over Ω —no time-domain, no Monte Carlo. The eigenvalue limit over-grants capacity; the gap widens with $d_F(A_{\text{sys}}(\rho))$.

D. Dynamic Braess as certificate disagreement

A reinforcement $\Delta w_{ij} > 0$ raises ν_c (Rayleigh monotonicity, $\partial \nu_c / \partial w_{ij} = (q_{c,i}/\sqrt{M_i} - q_{c,j}/\sqrt{M_j})^2 > 0$) yet raises $\kappa(V)$ and hence $\sup_\Omega \sigma_1$. The eigenvalue certificate improves while the transient certificate worsens: Braess is a sign disagreement between the two, to be resolved by certifying with γ_∞ , not ν_c .

VII. NUMERICAL STUDY: IEEE 39-BUS

We convert SGs at buses 30–39 to GFL units with realistic low-bandwidth PLLs and sweep ρ . The study reports: (A) $d_F(A_{\text{sys}})$ and γ_∞ rising with ρ ; (B) a penetration with $\alpha(A_{\text{sys}}) < 0$ (eigenvalue-safe) yet $\sup_\Omega \sigma_1 > \Gamma$; (C) optimal-forcing ψ_1 and optimal-response ϕ_1 maps localized at different buses; (D) the gap $\rho_{\text{eig}}^* - \rho_{\text{io}}^*$. Inter-area modes of the 39-bus system make the non-normal resonance pronounced and the input/output mismatch clearly separable. (Quantitative figures to be inserted from the authors' simulation.)

VIII. CONCLUSION

Modeling the inverter-dominated grid as a multi-agent system exposes that grid-following control makes its operator non-normal, so eigenvalue and voltage criteria certify the wrong object. The resolvent operator provides the correct, simulation-free certificate: a hidden amplification curve, non-modal weak nodes with input/output mismatch, a transient-aware hosting limit below the eigenvalue limit, and a resolvent reading of the Dynamic Braess Effect. Extending the operator to the non-symmetric directed Laplacian of high- R/X distribution feeders is the natural next step.

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